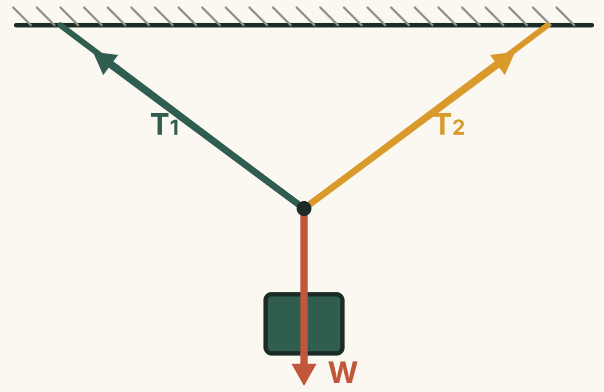
STATICS

# Forces in Balance

When several pulls act at one point yet nothing moves, they must add up to exactly nothing.




ESSENTIAL QUESTION

**What rule must concurrent forces obey for an object to stay perfectly still — or glide at constant speed?**

LESSON MAP

# Reach the rule in four moves.

**01**

### STATE

What “equilibrium” means — and when an object is in it.

**02**

### CONDITION

The single demand: the resultant of all forces is zero.

**03**

### CASES

Two forces line up; three forces close a triangle.

**04**

### SOLVE

Split into x and y, then read off the unknown forces.

KEEP THIS IN MIND

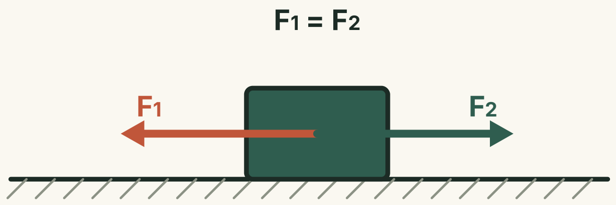
**Concurrent forces are pulls whose lines of action meet at one point.**

RECALL

# No net force means no change in motion.

Newton's first law: with zero resultant force, a body stays at rest or moves in a straight line at constant speed.

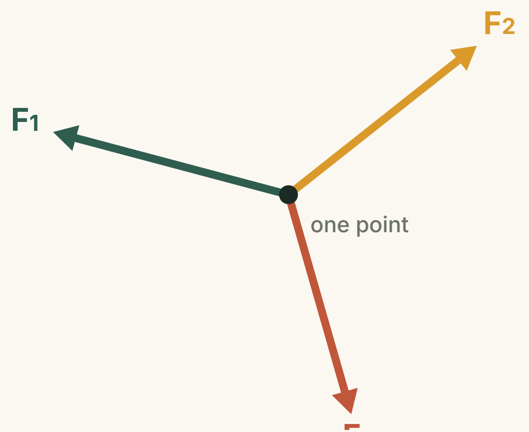
1. **Two equilibrium states**
   At rest, or in uniform straight-line motion — both have zero acceleration.
2. **Acceleration is the tell**
   Zero acceleration is exactly what balanced forces produce.



THE BIG IDEA

# Every pull is cancelled by the rest.

Slide the forces head-to-tail. In equilibrium the chain returns to where it started — the vector sum is the zero vector.




Concurrent forces share a point of action, so we treat the body as a single dot and just add the arrows.

THE BALANCE CONDITION

$$\sum \vec{F} = 0$$

CASE ONE

# Two forces: equal, opposite, in line.

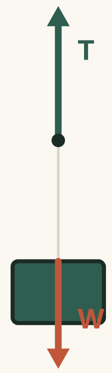
The simplest balance. A hanging weight feels just the rope's tension up and gravity down.

THE TWO-FORCE RULE

The forces must be equal in size, opposite in direction, and act along the same line.

IN BALANCE

$$\sum \vec{F} = 0$$



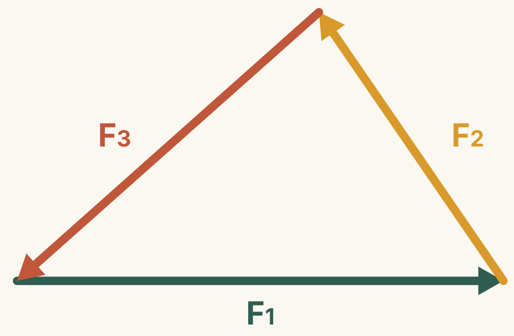
CASE TWO

# Three forces close a triangle.

Lay the three vectors head-to-tail. In equilibrium the last arrow lands exactly on the first — no gap.

1. **Any one balances the other two**
   One force equals the resultant of the rest, reversed.

2. **Their lines meet at a point**
   Three non-parallel balanced forces are always concurrent.



THE CONDITION

# The condition for equilibrium.

A vector equation is really two scalar ones. Pick axes and demand balance on each.

| | |
|---|---|
| $\sum F_x = 0$ | every force's x-components cancel |
| $\sum F_y = 0$ | every force's y-components cancel |

NO RESULTANT, SO NO ACCELERATION

$$\sum \vec{F} = 0$$

WHAT IT PROMISES

# Four things equilibrium guarantees.

### 01 · Zero resultant

All arrows, added tip-to-tail, return to the start.

### 02 · Static or uniform

Holds at rest or drifting at constant velocity.

### 03 · One balances the rest

Any single force equals the others' resultant, reversed.

### 04 · Concurrent lines

Three non-parallel balanced forces meet at a point.

ONE PICTURE TO HOLD $\sum F_x = 0 \quad \sum F_y = 0$

THE METHOD

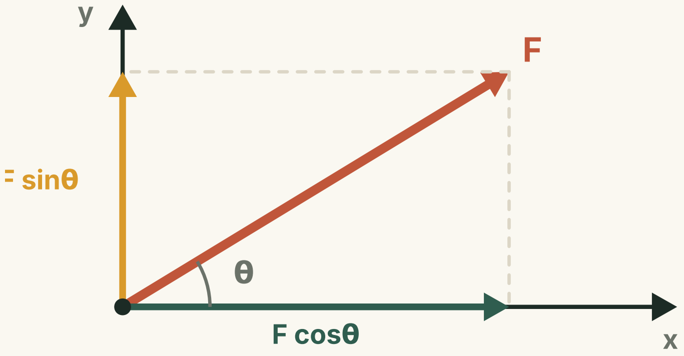
# Split each force onto two axes.

The reliable recipe: choose x and y, resolve every force, then set each column's sum to zero.

1. **Choose smart axes**
   Often along and across a surface, so awkward forces vanish.

2. **Resolve every force**
   Components $F_x = F\cos\theta, \quad F_y = F\sin\theta.$

3. **Set each sum to zero**
   Two equations, solve for the unknowns.



WORKED EXAMPLE

# A lamp held by two slanted ropes.

A 20 N lamp hangs from a knot, each rope at 30° above the horizontal. Find the tension in each rope.

1. **Balance the vertical**

   $$2T\sin 30^\circ = W$$

2. **Solve for the tension**

   $$T = \frac{W}{2\sin 30^\circ} = \frac{20}{2(0.5)} = 20\ \mathrm{N}$$

3. **Horizontal looks after itself**

   By symmetry the two horizontal pulls cancel.

THE SET-UP

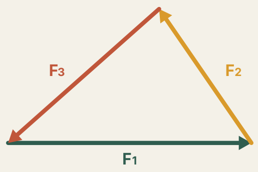
Two equal tensions, each lifting half the weight through their vertical component — symmetry does the rest.

SUMMARY

# Forces meet at a point in equilibrium only when their resultant is zero.

THE WHOLE RULE, ON TWO AXES

$$\sum F_x = 0 \quad \sum F_y = 0$$




Resolve every force onto x and y, balance each axis, and the unknown forces fall out.

**Balanced** is the cause; **no change in motion** is the effect.